Dr. Mohamed Yassin Laissy

Associate Professor, PhD in Civil Engineering (Structures)

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Professional Summary

Associate Professor with 20+ years of academic, research, and leadership experience. Accredited academic accreditation reviewer (NCAAA) and ABET program leader with a proven record in institutional and programmatic accreditation, internal auditing, curriculum and outcomes assessment, and educational governance. Experienced in building quality assurance systems, preparing self-study reports, and aligning academic programs with Vision 2030 and international standards. Research background in structural engineering, seismic design, sustainable concrete, and AI-integrated structural systems.

Key Competencies & Keywords

- Institutional accreditation (ETEC, NCAAA) and programmatic accreditation (ABET) leadership.
- Academic quality assurance, institutional effectiveness, and compliance monitoring.
- Policy development, governance frameworks, KPIs, and continuous improvement cycles.
- Curriculum design, program mapping, CLO–PLO assessment, and outcomes-based education.
- Internal auditing; quality specialist/practitioner certifications; risk and GRC practices.
- Stakeholder engagement, faculty development, and change management.
- Vision 2030 alignment; data-informed decision-making; AI in education and research.

Accreditation & Academic Quality Leadership

- Accredited Academic Accreditation Reviewer (ETEC, NCAAA), participating in hybrid and on-site reviews, including MSc Urban Design (KAU, 2022).
- Led Civil Engineering program readiness and documentation for ABET standards; contributed to university-wide NCAAA institutional accreditation and periodic program reviews.
- Chaired/served on committees preparing program self-study files, annual reports, KPIs dashboards, and improvement action plans.
- Established internal QA mechanisms (course reports, assessment plans, rubrics, and closing-the-loop practices) across Civil Engineering and broader college units.
- Delivered and organized training for faculty on NCAAA, ABET, assessment, and governance processes; supported audit readiness and evidence management.

Academic Appointments

- **Civil Engineering Department;** Associate Professor, University of Prince Mugrin (UPM), Saudi Arabia, *2025 – Present.*

- **Head of Civil Engineering Department;** Associate Professor, University of Prince Mugrin (UPM), Saudi Arabia, *2024 – 2025t.*
- **Head of Civil Engineering Department;** Assistant Professor, University of Prince Mugrin (UPM), Saudi Arabia**,** *2017 – 2024.*
- **Assistant Professor,** Department of Civil Engineering, University of Prince Mugrin (UPM), Saudi Arabia, *2016 – 2017.*
- **Assistant Professor,** Department of Architectural Engineering, Omar Al-Mukhtar University (OMU), Libya *2014 – 2015.*
- **Assistant Professor,** Department of Civil Engineering, El-Zawiya University (ZU), Libya *2013 – 2014.*
- **Assistant Professor,** Department of Civil Engineering, Modern University for Technology & Information (MTI), Cairo, Egypt**,** *2012 – 2013.*
- **PhD Scholar,** Faculty of Engineering, Cairo University, Egypt, *2005 – 2010.*
- **Civil Engineer** (Government and industry roles), Egypt *1993 – 2012.*

Education

- **PhD, Civil Engineering (Structures), Cairo University, Egypt** (2010)
- Thesis: Finite Element Analysis of Blast and Low-Velocity Impact on Sandwich Panels with Multiple Suppressive Cores
- **MSc, Civil Engineering (Structures), Ain Shams University, Egypt** (2002)
- **BSc, Civil Engineering, Cairo, Egypt** (1993)

Certifications & Professional Development (Selected)

- Academic Accreditation Reviewer (ETEC, NCAAA), Education & Training Evaluation Commission, 2022.
- Internal Auditor, Quality Specialist, Quality Practitioner – Saudi Society for Quality (2019); ongoing QA workshops at UPM 2019.
- Certified GRC Professional (GRCP) & Certified GRC Auditor (GRCA) – OCEG/GRC Certify, 2025.
- Institutional Governance Program (20 hours), 2024.
- Saudi Building Code SBC 301 Requirements – Saudi Council of Engineers, 2023.
- ACI University: Using ACI 318-14 for Design of Shear Walls & Earthquake-Resistant Shear Walls, 2023.
- AI & ML: Building Data Science Solutions – MIT Professional Education (80 hours), 2025.
- Training in AI in education, project management, KPIs, assessment, and e-learning (2021–2025)

Administrative, Governance & Service

- **UPM:** Head of Civil Engineering Department Council (2017–2025); Head of the college curriculum review committee (2024–present); Member of the University Council (2016–Present); Head/Member of numerous committees, including Operational Budget, Appeals, QA & Accreditation (2016–2018, ongoing).
- **OMU:** Head of Scientific Committee; Member of Faculty Council; Head of Examination Control and Academic Committees; University QA & Accreditation Committee (2013–2015).
- **MTI:** Member of Faculty Council; Member of examination control (multiple levels); QA unit, questionnaires and competency evaluation committees (2012–2013).

Teaching Experience – Selected Undergraduate Courses,

- Construction Field Safety & Hazard Management (UPM).
- Durability, Evaluation & Repair of Concrete Structures (UPM).
- Capstone Projects I & II; Structural Analysis; Strength of Materials; Statics & Dynamics (UPM).
- Structural Analysis I–III; Reinforced Concrete; Mechanics of Solids; Building Construction I–II (OMU, ZU, MTI, El-Oubour).

Research Interests

AI-integrated structural systems; seismic-resistant design; sustainable concrete; blast/impact and projectile penetration; structural analysis and retrofitting.

Publications (Selected Peer-Reviewed)

- Laissy, M.Y., Rashed, H.F., Helal, K., Alfadil, O.A. (2026). Culturally inspired acoustic retrofit for high-density pilgrimage corridors: a pilot physical model study and reproducibility-oriented validation protocol. Frontiers in Built Environment, 12, Article 1959288. https://doi.org/10.3389/fbuil.2026.1959288
- Jassam, T., Rong Ying, L., Ri Teng, L., Jia Hao, C., Allujami, H., Laissy, M. (2025). Producing eco-friendly concrete using various recycled materials for sustainable construction. European Journal of Environmental and Civil Engineering, 29(14), 3145–3180. https://doi.org/10.1080/19648189.2025.2512844.
- Laissy, M.Y., Alkhatib, M.-M., Kussayer, A.F., Ismail, H., Rashed, H.F. (2025). Sustainable Restoration and Preservation of Makhet Castle of the Hejaz Railway, Both Structurally and Culturally. Lecture Notes in Civil Engineering, Vol. 556, Springer, pp. 106–114. https://doi.org/10.1007/978-981-97-8712-8_14.
- Laissy, M.Y., Rashed, H.F. (2025). 3D Printing Technology for Construction: A Structural Shift in Building Infrastructure. Proceedings of the ICSDI 2024, Volume 3, Springer, pp. 135–142. https://doi.org/10.1007/978-981-97-8345-8_17.
- Laissy, M., Belbol, B., Boshi, O., Eldeiasti, A. (2024). AI Analysis of the Thermal Effects on Reinforced Concrete Buildings with Floating Columns. Engineering, Technology & Applied Science Research, 14(5), 16154–16159. https://doi.org/10.48084/etasr.8160.
- Laissy, M., Al-Turkmani, Z., Atarji, O., Saud, A. (2023). A Comparative Study of the Seismic Response of Different Concrete Slab Systems for a Multistory Building in Madinah. Engineering, Technology & Applied Science Research, 13(5), 11715–11720. https://doi.org/10.48084/etasr.6047.
- Laissy, M.Y. (2022). Effect of Different Types of Bracing System and Shear Wall on the Seismic Response of RC Buildings Resting on Sloped Terrain. Civil Engineering Journal, 8(9), 1958–1976. https://doi.org/10.28991/CEJ-2022-08-09-014.
- Elhozayen, A., Laissy, M.Y., Attia, W.A. (2019). Investigation of High-Velocity Projectile Penetrating Concrete Blocks Reinforced by Layers of High Toughness and Energy Absorption Materials. Civil Engineering Journal, 5(7), 1518–1532. https://doi.org/10.28991/cej-2019-03091349.

- El Ouni, M.H., Laissy, M.Y., Ismaeil, M., Ben Kahla, N. (2018). Effect of Shear Walls on the Active Vibration Control of Buildings. Buildings, 8(11), Article 164. https://doi.org/10.3390/buildings8110164.
- Youssef, R., Attia, W., Laissy, M. (2017). Numerical Simulation of Concrete Slabs Strengthened with PTFE Sheets Subjected to Blast Load. MATEC Web of Conferences, 120, Article 04005. https://doi.org/10.1051/matecconf/201712004005.
- Tawadrous, R.I., Attia, W.A., Laissy, M.Y. (2016). Using Ceramic Plates as Shielding for Concrete Blocks Against Projectile Penetration. HBRC Journal, 12(3), 263–271. https://doi.org/10.1016/j.hbrcj.2014.11.011.
- Tawadrous, R., Attia, W., Laissy, M. (2014). Investigating Projectile Penetration of Concrete Blocks Reinforced by Ceramics. WIT Transactions on the Built Environment, 141, Structures Under Shock and Impact XIII. https://doi.org/10.2495/SUSI140101.

Conference Presentations

- **SusTAIN 2026:** Valorization of Palm Waste in Cementitious Concrete-Block Envelopes for Sustainable Thermal Insulation in Hot-Arid Buildings. The 1st International Conference on Technology and Engineering Innovation for Sustainability, October 6–8, 2026 (Madinah, Saudi Arabia).
- **ICSDI 2026:** AI-Driven Hazard Risk Assessment in Construction: Integrating Multi-Site Field Reports with Machine Learning. The 3rd International Conference on Sustainability: Developments and Innovations, February 8–12, 2026 (Riyadh, Saudi Arabia).
- **ICSDI 2026:** Thermal Evaluation of Date Seed-Filled Concrete Blocks for Sustainable Insulation in Hot Climates. Co-authored conference paper, published in Springer, Lecture Notes in Civil Engineering, Vol. 886 (Riyadh, Saudi Arabia).
- **ICASF 2024:** AI Tool for Sustainable Project Management Construction (SPMC). The 2nd International Conference on Advancing Sustainable Futures, December 11–12, 2024 (Abu Dhabi, UAE).
- **ICSDI 2024:** 3D Printing Technology for Construction: A Structural Shift in Building Infrastructure; Sustainable Restoration and Preservation of Makhet Castle of the Hejaz Railway, Both Structurally and Culturally. The 2nd International Conference on Sustainability: Developments and Innovations, February 18–22, 2024 (Riyadh, Saudi Arabia).
- **WIT CMEM XIX (2019):** Numerical Modeling of High-Velocity Projectile Penetrating Concrete Blocks Reinforced by Teflon Sheets. The 19th International Conference on Computational Methods and Experimental Measurements, June 3–5, 2019 (Seville, Spain).
- **ASCMCES-17 (2017):** Numerical Simulation of Concrete Slabs Strengthened with PTFE Sheets Subjected to Blast Load. International Conference on Advances in Sustainable Construction Materials & Civil Engineering Systems, April 18–20, 2017 (Sharjah, UAE).
- **WIT SUSI XIII (2014):** Investigating Projectile Penetration of Concrete Blocks Reinforced by Ceramics. The 13th International Conference on Structures Under Shock and Impact, June 3–5, 2014 (New Forest, UK).
- **WIT SUSI XII (2012):** Investigating Projectile Penetration of Sandwich Panels with Multiple Suppressive Cores. The 12th International Conference on Structures Under Shock and Impact, September 4–6, 2012 (Kos, Greece).

- **MTC ICCAE-8 (2010):** A Review of the Blast Effects on Sandwich Panels with Suppressive Cores; Experimental and Numerical Simulation of Projectile Penetration in Concrete. The 8th International Conference on Civil and Architecture Engineering, May 25–27, 2010 (Cairo, Egypt).

Invited Panel Discussions

- **SusTAIN 2026:** Invited Panelist, "AI and Emerging Technologies: Shaping the Sustainable Cities and Societies of the Future." The 1st International Conference on Technology and Engineering Innovation for Sustainability, October 8, 2026 (Madinah, Saudi Arabia).

Postgraduate Supervision

- **PhD:** High-Velocity Projectile Penetration in Concrete Blocks Reinforced by Toughness/ Energy-Absorbing Layers (Cairo University, 2018).
- **MSc:** Investigating Projectile Penetration of Concrete Block Reinforced by Aluminum (Cairo University, 2017).
- **MSc**: Investigating Projectile Penetration of Concrete Block Reinforced by Ceramics (Cairo University, 2013).
- **MSc**: Numerical simulation of concrete slabs strengthened with PTFE under blast load (Cairo University).

Funded Research & Grants

- Preliminarily accepted: Development and Validation of Predictive AI & IoT System for Monitoring and Forecasting Air Quality (RDIA, KSA) – aligned with Vision 2030 (SDG-11, SDG-13).

Professional Memberships

- ASCE – American Society of Civil Engineers (2018)
- ACI – American Concrete Institute (2018)
- Saudi Council of Engineers (2022–Present)
- Saudi Society for Quality (2018–Present)
- Egyptian Syndicate of Engineers (1993–Present)

References

- **Prof. Walid Abdellatif Attia** – Professor of Structural Analysis & Mechanics, Cairo University | waattia@gmail.com
- **Dr. Fouad Dehlawi** – Associate Professor, Faculty of Engineering, King Abdulaziz University | fdehlawi@gmail.com
- **Dr. Mohamed Ezz Eldeen** – Head of Civil Engineering, Modern University for Technology & Information.
- **Dr. Aisha Bakar** – Head of Architectural Engineering, Omar Al-Mukhtar University | aishabakar@yahoo.com

Mohamed Laissy

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